

Contribution of Subsurface Flow from the Upper Slopes of Forested Watersheds to Channel Flow¹

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ABSTRACT

Overland flow and subsurface flow from 0.054- and 0.089-ha plots on the upper slopes of forested watersheds were intercepted and measured in trenches located at the bases of the plots. Channel flow was measured in flumes at the watershed outlets and intensity, duration, and amount of rainfall were measured by gages on the watersheds. Over a 3-year period, 115 storms produced little overland and subsurface flow above the B-horizon. Subsurface flow over an impermeable clay layer was as much as 90% of rainfall during a calendar quarter. Thirty-six of the storms produced subsurface flow on one or both plots. For these storms, subsurface flow from the upper third of the watersheds was equivalent to 26% of the channel flow on one watershed and to 27% on the other. For 27 of the 36 storms, subsurface flow from the plots peaked sooner than channel flow. Twelve storms produced subsurface flow within 20 min of the onset of rainfall, and 15 storms produced channel flow within the same time. Subsurface flow often began shortly after rainfall began even when there was neither saturation at the point of outflow nor high antecedent soil moisture. Interconnected channels through the soil formed by decayed roots and animal burrows are the best explanation for the quick subsurface flow responses. The similarity in timing of subsurface and channel flows indicates that subsurface flow from upper slopes can contribute significantly to storm hydrographs in forested areas where permeable surface soils overlie an impermeable stratum.

Additional Index Words: lateral flow, soil moisture, hydrology, runoff, streamflow.

INVESTIGATORS DISAGREE about the timing, mechanics, and magnitude of subsurface flows from upper slopes. Some deny that subsurface flow contributes significantly to runoff (Dunne and Black, 1970; Freeze, 1972); others maintain that only subsurface flow from near the stream channel contributes to storm hydrographs (Hewlett and Hibbert, 1967; Hewlett and Nutter, 1970; Nutter 1973). Gaiser (1952), Whipkey (1967), and Aubertin (1971), however, assert that interconnected macrochannels formed by roots and animal burrows can provide the means for rapid subsurface flow from upper slopes to stream channels. This paper examines the contribution of subsurface flow from upper slopes to channel flow and discusses flow timing in relation to rainfall and antecedent soil moisture for two forested watersheds in northern Mississippi.

METHODS

Study Area

The study area is typical of much of the terrain in the upper Coastal Plain. The two watersheds studied support mature short-leaf pines (*Pinus echinata*) and mixed hardwoods. The upper slopes have average gradients of 30% and are composed of two

highly permeable Coastal Plain soils, McLaurin and Smithdale sandy loams. Wilcox sandy loam, a poorly drained soil with very low permeability, dominates the lower slopes. A dense stratum of kaolinite clay lies 6 m below the surface at the watershed divides but converges to within < 1 m of the surface at midslope.

Mean annual precipitation on the study area is 135 cm; 80 cm fall from November through April. The lowest monthly rainfall average is 8 cm in October; the highest is 16 cm in March.

Installations and Measurements

Study plots were established on the upper slopes of the adjacent 1.86- and 1.62-ha watersheds. The surface drainage area of plot 1, located on the larger watershed, was 0.054 ha; that for plot 2 was 0.089 ha. The subsurface drainage areas, defined by augering to the clay layer and preparing a contour map of its surface, were 0.045 ha on plot 1 and 0.068 ha on plot 2. Average depth to the clay is 1.8 m on plot 1 and 2.1 m on plot 2.

Trenches, located about 75 m below the watershed divides and 125 m above the watershed outlets, served as the lower boundaries of the plots. They were 12.2 m long and extended into the clay; two gravel-filled troughs were constructed inside of each (Fig. 1). One trough intercepted subsurface flow from the A-horizon; the other intercepted flow from between the A/B-horizon interface and the clay. Subsurface flow at the two levels was diverted through sheet metal troughs into 0.305 m HS-flumes equipped with stage recorders. A strip of sheet metal was inserted between the litter layer and mineral soil at the lip of each trench to divert overland flow to 0.153 m H-flumes for measurement. The collection troughs were driven into the trench faces to prevent water from seeping around one trough to the one below. The trenches and diversion troughs were covered to prevent the direct inflow of precipitation. Before the trenches were constructed, subsurface flow emerged through holes about 5 to 10 cm in diameter that were 15 to 30 m below the plots. From that point flow continued down-slope in well-defined intermittent stream channels. The trenches interrupted the natural pathway of subsurface flow, but the water they intercepted moved readily through the collection system and became part of channel flow. Channel flow, water from all sources (including the plots) that reached the watershed outlets, was measured in 0.915 m H-flumes constructed in the natural channels. Concrete cut-off walls beneath the flume approach sections extend into clay to minimize underflow.

Rainfall on the two watersheds was measured by a network of five gages. A centrally located gage gave rainfall durations and intensities.

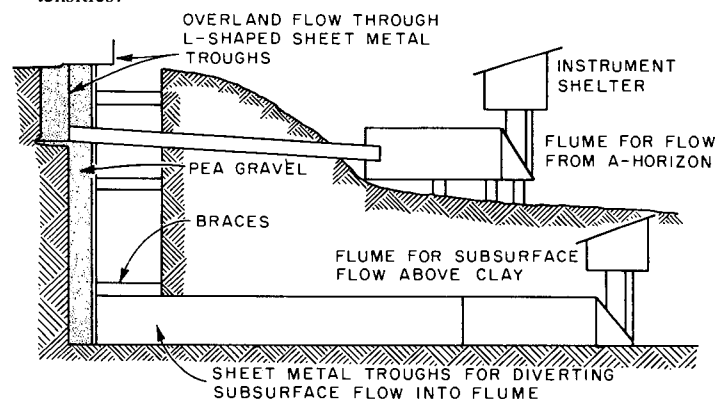


Fig. 1—Diagram of interceptor and diversion trenches with installations for measuring overland flow, flow from A-horizon, and subsurface flow above clay layer.

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Analyses

Rainfall and subsurface flow volumes above the clay layer were totaled by calendar quarter for the 115 storms (rainfall events that produced subsurface flow on one or both plots) that occurred from 1971 through 1973. The storms were used to compare subsurface flow volumes with total rainfall by calendar quarters during the three years. Thirty-six storms that produced subsurface flow on both plots were used to compare: (i) subsurface flow volumes measured at the plots with channel flow volumes measured at the watershed outlets, (ii) time to beginning of subsurface flow with time to beginning of channel flow, and (iii) time to subsurface flow peaks with time to channel flow peaks. Regressions were developed using a combinatorial screening technique to relate the beginning and peaks of subsurface and channel flows to (i) maximum 10-min rainfall intensity, (ii) time to midpoint of maximum rainfall intensity, (iii) rainfall amount, and (iv) antecedent rainfall. An additional variable, antecedent soil moisture, was included along with the rainfall variables in a regression analysis of 26 storms for which soil moisture data were available.

RESULTS AND DISCUSSION

Subsurface' Flow as a Percentage of Rainfall

Overland flow and subsurface flow above the B-horizon were negligible during the study period; infiltration rates on the upper slopes were high enough to accommodate even the most intense storms. Subsurface flow between the A/B horizon interface and the clay layer, however, often was a large percentage of rainfall, especially during wet periods (Table 1). During quarter 1 of 1973, 88% of the 75.4 cm of total rain emerged as subsurface flow on plot 1, and 92% emerged on plot 2. A storm that delivered over 25 cm of rain in 42 hours produced about one-third of the subsurface flow measured during that quarter. During quarter 2 of 1973, subsurface flow accounted for 51% of the total rainfall on plot 1 and 37% on plot 2. During periods of lower rainfall a smaller percentage emerged from the plots as subsurface flow because most of the water went to evapotranspiration and soil storage. One hundred thirty-two cm of rain fell in 1971, which is slightly less than average; only 5.7% of the rain emerged as subsurface flow. In 1972, rainfall was 20 cm above average and 11.5% emerged as subsurface flow. In 1973, rainfall exceeded the average by 44 cm and 47.7% of the rain emerged as subsurface flow.

Table 1—Subsurface flow as a percentage of total rainfall for 115 storms by calendar quarters and years.

Year	Total rainfall	Subsurface flow	
		Plot 1	Plot 2
	cm		%
1971			
Quarter 1	45.87	11.3	14.0
2	22.35	2.1	1.6
3	43.18	1.8	1.6
4	20.32	1.6	0.4
Annual	131.57	5.1	5.7
1972			
Quarter 1	31.75	12.0	12.2
2	36.32	0.7	1.3
3	33.27	0.9	3.2
4	52.58	17.8	23.4
Annual	153.92	8.8	11.5
1973			
Quarter 1	75.44	88.2	91.6
2	39.12	50.8	36.9
3	16.51	0.4	0.4
4	47.75	3.2	3.8
Annual	178.82	49.2	47.7

Comparing Subsurface and Channel Flow Volumes

Subsurface flow from the upper slopes often was a significant proportion of total flow from the catchments. Total subsurface flow from plot 1 for 36 storms over the 3-year period was 82.9 area cm compared with 91.0 area cm of channel flow. Subsurface flow from plot 2 was 82.6 area cm compared with 92.7 area cm of channel flow. Assuming the plots represent the upper one-third of the watersheds, subsurface flow from the upper slopes is equivalent to 27% of channel flow on watershed 1 and 26% on watershed 2 for these 36 storms. Betson (1964) and Hewlett (1961) contend that most channel flow comes from small areas adjacent to the stream, and Freeze (1972) concludes that overland flow from near the stream channel dominates the storm hydrograph on well-vegetated basins. The results of the current study, however, demonstrate that subsurface flows from upper slopes can be important contributors to channel flow volumes.

Timing of Flows in Response to Rainfall

The elapsed time from the onset of rainfall to the starts and peaks of flows showed, as Kirkby and Chorley (1967) found, that subsurface flow often begins soon enough to contribute to channel flow peaks. A distribution of flow initiation times by 20-min intervals shows a high proportion of storms produced subsurface flow at the plots soon after rain began. Twelve of the 36 storms produced subsurface flow within 20 min of the beginning of rainfall (Fig. 2). Fifteen storms, including 11 of the 12 that had quick subsurface flow, caused a rise in channel flow within the same time.

The average time lag between the onset of rainfall and a rise in the channel hydrograph was 93 min for the 36 storms. The delay was caused by the few storms that began gradually and continued at low intensities, sometimes for hours, before producing stormflows.

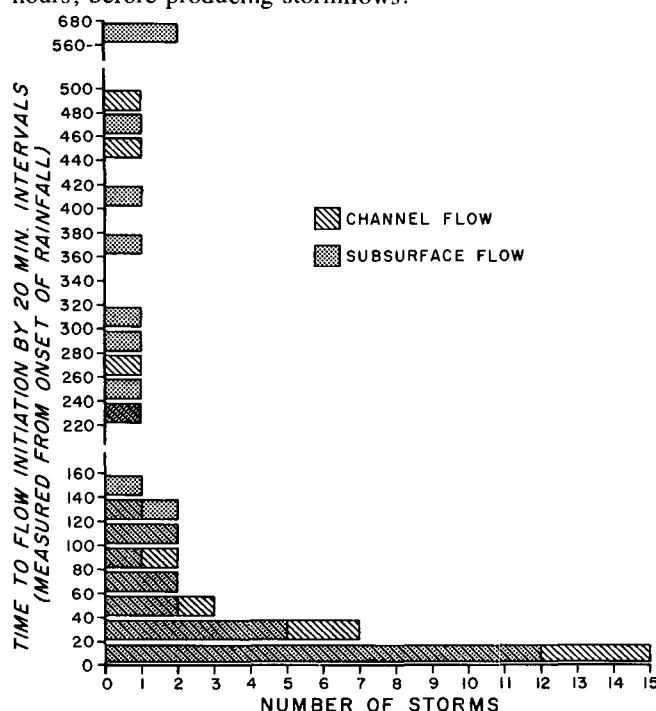


Fig. 2—Distribution of times to initiation of subsurface and channel flows for 36 storms.

The quick response of subsurface flow to rainfall cannot be explained solely on the basis of saturated flow through the main soil mass. Measured saturated hydraulic conductivities of the highly permeable surface soils ranged up to 64 cm per hour, but even these fast rates are not sufficient to account for peaks occurring < 1 hour from the onset of rainfall. Translatory flow (Hewlett and Hibbert, 1967) the process by which soil water near the stream channel is pushed out of the soil by rain infiltrating upslope, does not explain the quick responses because it requires a saturated zone near the point of outflow. On 13 Aug. 1974, 10 tensiometers, 2 at each 0.305 m interval down to the clay, were installed immediately above the trench on plot 1. Five storms produced quick subsurface flows during the following 3 months. Three of the storms produced a peak within 1 hour of the onset of rainfall. At no time during this period was there a saturated zone above the trench. Timing of sub-

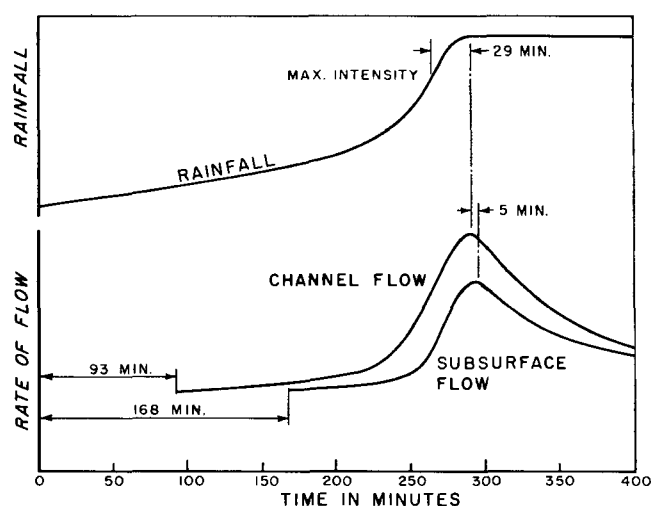


Fig. 3—Hydrographs showing average time from onset of rainfall (time zero) to maximum rainfall and times from beginning of flows to flow peaks for 36 storms. (The vertical scales are not quantitative.)

Before subsurface flow can occur in macrochannels, the channels must be open to the atmosphere and a positive head of water must exist at the openings. Such conditions are common in a forest environment. Depressions formed by uprooted trees or decayed stumps provide a place for water to concentrate. Decayed roots spreading outward from the depressions provide natural pathways for water to flow freely through the soil toward the stream channels. Stemflow accumulating at the bases of trees is another possible source of significant quantities of subsurface flow.

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